

Class-Aware Data Augmentation for Long-Tailed Object Detection on the BadODD Dataset

Image-level oversampling and targeted bounding-box-safe augmentation on a fixed YOLO11s detector

MD SAIDUL ISLAM (22-49719-3) | Mehedi Hasan Raj (22-49733-3) | Tanvir Hassan Tonmoy (22-49897-3)
Supervisor: Tanvir Ahmed | Department of Computer Science and Engineering, American International University-Bangladesh



Introduction

Object detection is essential for autonomous driving, yet Bangladeshi roads present unique challenges due to dense, heterogeneous traffic and uncommon vehicle types such as auto-rickshaws, three-wheelers, and cart-vehicles. The BadODD dataset captures these conditions across nine districts, with day/night images and thirteen annotated classes. However, its highly imbalanced class distribution causes detectors to perform well on frequent categories while underperforming on rare but safety-critical classes. Existing studies have focused on detector architectures, leaving the impact of data-centric strategies for addressing class imbalance largely unexplored.

RESEARCH QUESTION

Can class-aware image sampling and targeted augmentation improve detection of under-represented categories in BadODD while maintaining overall performance?

Methods

Two data-centric interventions are applied to a fixed YOLO11s detector initialised from the COCO-pretrained checkpoint. Neither modifies the architecture, the loss function or the optimisation procedure; the detector is an instrument, not a contribution.

A. Class-aware image-level oversampling

Oversampling is applied to whole images rather than by duplicating annotations, since an object is embedded in a scene. A per-class repeat factor uses square-root dampening with a hard ceiling:

$$r_c = \min(r_{\max}, \sqrt{n_{\text{med}} / n_c}), r_{\text{img}} = \text{ceil}(\max r_c \text{ over minority classes in the image}), r_{\max} = 4$$

Applied to this dataset, 70 of 5,967 training images (1.17 %) contain a minority-group class. All three uncapped factors (8.86, 9.47, 8.06) exceed the ceiling, so every selected image receives a factor of exactly 4. Validation and test manifests are left unmodified.

B. Targeted bounding-box-safe augmentation

Two variants per selected image are generated offline with Albumentations; boxes are transformed jointly with the image, re-clipped, and discarded below 30 % of original visible area or 16 px. The transform set is deliberately conservative: horizontal flip ($p = 0.5$), scale jitter ± 20 % and translation ± 10 % ($p = 0.6$), rotation $\pm 7^\circ$ ($p = 0.5$), brightness/contrast ± 20 % ($p = 0.5$), hue/saturation ($p = 0.25$), light blur and noise ($p = 0.1$). Strong degradation was avoided because rare objects are small. This produced 140 augmented images and 70 exact duplicates; a random sample was rendered with overlaid boxes and visually verified before training.

Setup: YOLO11s (9.43 M params), 512×512 , 25 epochs, batch 16, SGD ($\text{lr}_0 = 0.01$, momentum 0.937), seed 42, single NVIDIA Tesla T4 (15.6 GB). Early stopping disabled. Test split held out entirely until final evaluation.

Confound disclosed: the class-aware configuration trains on 6,177 images per epoch against the baseline's 5,967 — an expansion factor of 1.035, and ≈ 3.5 % more optimisation steps. The step budget was not matched.

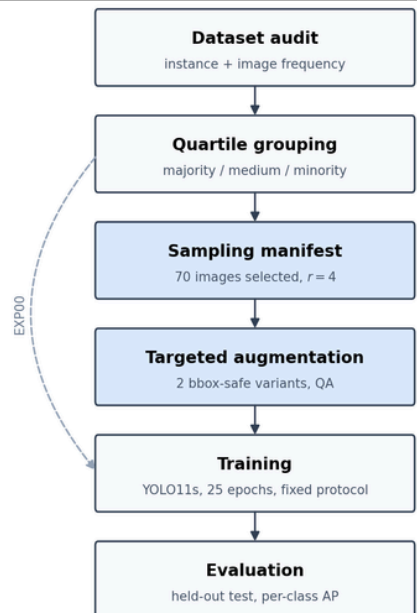


Fig. 2. Pipeline of the proposed method. The baseline configuration bypasses the sampling and augmentation stages and trains directly on the unmodified manifest.

Novelty

We propose a BadODD-specific class-aware data training strategy that combines image-level oversampling with targeted bounding-box-safe augmentation for under-represented classes, while keeping the YOLO11s detector architecture, loss function, and optimization procedure fixed.

Dataset and Problem Analysis

Reported versus observed

Direct enumeration of the Zenodo release gives 10,032 images and 80,036 annotated objects against a reported 9,825 and 78,943.

Two classes account for most of the gap: the obtained training split holds 21 train and 29 wheelchair instances against a reported 1 and 2. A repartition of the same data cannot explain this, so both sets of values are reported. No class-name mapping file was distributed; names were recovered and verified against the reported class-frequency table.

Long-tail structure

The training split contains 47,514 objects across 13 classes.

Imbalance ratio = 869.9; coefficient of variation = 1.408.

Instance and image frequency diverge sharply: person appears 18,268 times across 4,807 images (3.80 per image), whereas each of the three rarest appears about once per image. Image-level imbalance is therefore milder — 253.0 against 869.9.

Data-driven grouping (fixed before training)

Quartiles of positive training counts: $Q_1 = 144$, Median = 1,885, $Q_3 = 3,786$. Majority — person, auto-rickshaw, three-wheeler Medium — car, motorbike, truck, bus, bicycle, priority-vehicle, cart-vehicle. Minority — wheelchair, construction-vehicle, train (test support 28, 11, 16).

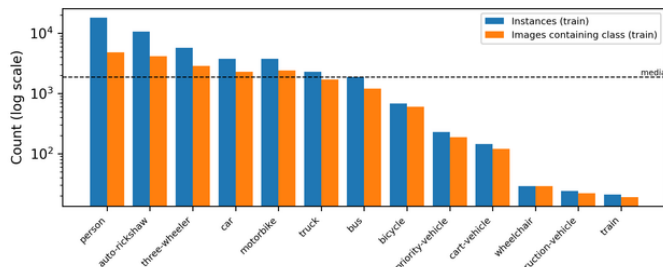


Fig. 1. BadODD training-split class distribution on a logarithmic scale, showing instance frequency and the number of images containing each class. The dashed line marks the median class count used in the repeat-factor computation.

TABLE 1. BadODD statistics: reported versus observed

Property	Reported	Observed
Total images	9,825	10,032
Total annotated objects	78,943	80,036
Classes	13	13
Train / val / test images	5,896 / 1,964 / 1,965	5,967 / 2,032 / 2,033
Train inst., constr.-veh.	23	24
Train inst., wheelchair	2	29
Train inst., train	1	21
Test inst., train	0	16

Results

All values were obtained on the held-out test split of 2,033 images, evaluated once after both runs completed. Two configurations were trained: EXP00 (baseline, unmodified manifest) and EXP03 (both components combined).

Overall: The class-aware configuration recorded a higher $\text{mAP}@0.50$ (62.86 vs. 60.78, +2.08 points) and $\text{mAP}@0.50:0.95$ (40.24 vs. 38.58, +1.67). Precision fell by 2.10 points and recall by 1.06. The direction of change is not uniform across the four metrics.

By class group: The difference is concentrated where the interventions were applied. At $\text{AP}@0.50$ the majority group changed by +0.03, medium by +0.93, and minority by +6.83; at $\text{AP}@0.50:0.95$, +0.17, +0.15 and +6.69 respectively. Among the seven classes with more than 600 training instances, every change at $\text{AP}@0.50$ fell within $[-0.65, +0.66]$ — no meaningful cost to frequent categories at this intervention strength.

TABLE 2. Overall detection performance (test split)

Configuration	P	R	mAP50	mAP50:95
EXP00 baseline	75.41	54.88	60.78	38.58
EXP03 class-aware	73.30	53.82	62.86	40.24
Δ	-2.10	-1.06	+2.08	+1.67

TABLE 3. Group-level mean average precision

Group	n	mAP50 base	mAP50 prop.	Δ
Majority	3	79.05	79.08	+0.03
Medium	7	58.54	59.47	+0.93
Minority	3	47.74	54.56	+6.83

Minority = wheelchair, construction-vehicle, train (combined test support 55). At $\text{AP}@0.50:0.95$: +0.17, +0.15 and +6.69.

Findings and Qualifications

Per class: Eight of thirteen classes improved at $\text{AP}@0.50$. Largest increases: construction-vehicle +11.89 (0.17 $\rightarrow$ 12.06), wheelchair +6.80, priority-vehicle +5.56. Largest decreases: truck -0.65 , car -0.41 , three-wheeler -0.27 .

Qualifications: Construction-vehicle improved to 12.06 $\text{AP}@0.50$, but remains difficult. Priority-vehicle also gained +5.56, suggesting run-to-run variation.

Efficiency: Same model size (9.43M parameters, 19.15 MB); latency remained nearly unchanged (22.01 vs. 21.48 ms). Training increased slightly (67.3 $\rightarrow$ 69.7 min).

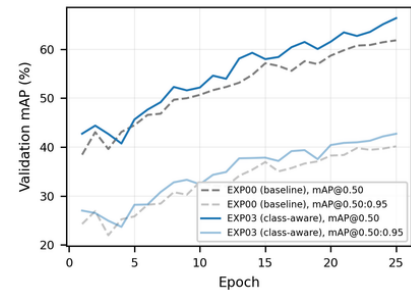


Fig. 3. Validation mAP across the 25-epoch budget or both configurations.



Fig. 4. Qualitative comparison on test images containing minority-group objects. Left column: baseline configuration. Right column: class-aware configuration. Detections rendered at a confidence threshold of 0.25.

Conclusion

The results are consistent with the hypothesis but do not establish it: a single seed provides no estimate of run-to-run variation, and untargeted categories moved by comparable amounts, so the null hypothesis cannot be rejected. Only the baseline and the combined configuration were trained, so the two components cannot be separated.

Next steps: multi-seed replication, separate runs for each component, and comparison against loss-reweighting methods.

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